

Jaeseung Kim

Undergraduate Researcher in Medical Imaging and Computational MRI
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RESEARCH INTERESTS

Computational MRI | Medical image learning | Quantitative imaging | Neuroimaging | Efficient deep learning

EDUCATION

Yonsei University			Expected Feb 2027
B.S. Candidate, Biomedical Engineering (Mirae Campus) Second Major, Electrical & Electronic Engineering (Sinchon Campus)			
GPA 4.03 / 4.30	4.50 Scale 4.18 / 4.50	Percentile 97.90 / 100	Department Rank 1 / 49
Selected Coursework	Image Processing and Applications; Human Physiology; Human Anatomy; Computer Programming; Digital Signal Processing; Signals and Systems; Electromagnetics		
Recent Coursework	Deep Learning Laboratory; Public Health and Medical Big Data Analysis; Bioelectronics		

RESEARCH EXPERIENCE

Advanced Image Processing Techniques Lab, Yonsei University	Dec 2023 - Feb 2026
Undergraduate Researcher Advisor: Prof. Joon Yul Choi Mirae Campus	
- Conducted 3D medical-imaging research spanning preprocessing, model development, quantitative evaluation, and statistical analysis. - Built reproducible volumetric imaging pipelines for registration, masking, resampling, quality control, and downstream analysis. - Investigated variability across MRI sequences, scanners, vendors, and acquisition parameters, emphasizing robust validation and harmonization. - Performed voxel-wise group analyses and interpreted quantitative imaging measures using SPM12-based workflows. - Prepared two first-author journal manuscripts; project-specific methods and results are withheld before submission at the advisor's request.	
Bio-Metamaterials Lab, Yonsei University	Jul 2022 - Aug 2022
Undergraduate Researcher Advisor: Prof. Seung-Ho Choi Mirae Campus	
- Supported the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials. - Developed a prototype tensile-testing device and implemented a MATLAB-based user interface for measurement verification.	

PUBLICATIONS, PRESENTATIONS, AND RESEARCH OUTPUTS

- Two first-author journal manuscripts in preparation in medical imaging; project-specific details withheld before submission.
- Oral Presentation and Outstanding Paper Award (Oral Presentation Division), Korean Society of Medical & Biological Engineering 2025 Fall Conference, Nov 2025.
- Poster accepted, NEURO2026, Jul 2026.

ENTREPRENEURSHIP AND LEADERSHIP

CareProofRecord - AI-Generated Medical Explanation Video Platform

Jun 2025 - Apr 2026

Project Lead

- Planned and developed an AI service intended to reduce information-transfer errors during surgical and clinical explanations and improve patient understanding.
- Designed a video-generation service architecture intended to mitigate hallucination risk and preserve long-context consistency.
- Received KRW 1,000,000 in support after review by the university entrepreneurship center in Sep 2025.
- Assessed potential clinical use cases and demand with prospective users, including regional medical centers.

SCHOLARSHIPS AND HONORS

- National Science & Engineering Scholarship, Korea Student Aid Foundation - full tuition, Spring/Fall 2023 and Spring 2026
 - Han Kyung-beom Scholarship, Yonsei University - full tuition, Spring/Fall 2022
 - Jinri Scholarship for Academic Excellence, Yonsei University, Fall 2021
 - Academic Achievement Awards, Yonsei University, 2021-2023
- Excellence Award, Yonsei University Algorithm Competition, 2022
 - **Grand Prize**, ASC Programming Competition, 2021
 - **Grand Prize**, MEDICI+ Startup Competition, 2021
 - **Honorable Mention**, WAVE-LAB Startup Competition, 2025
 - Encouragement Award, Post Corona-19 Startup Challenge, 2022

TECHNICAL SKILLS

Programming	Python; MATLAB; Bash; Linux
Medical Imaging	FSL; ANTs; SPM12; registration; masking; resampling; VBM; voxel-wise group analysis
Research Methods	3D medical-image preprocessing; quantitative evaluation; image-to-image regression; statistical interpretation